



STEM SMART Single Maths 13 - Using Integration >

Integration and Area

A-level Maths Topic Summaries - Calculus

Subject & topics: Maths | Calculus | Integration Stage & difficulty: A Level P2

Fill in the blanks below to complete these summary notes on integration and area.

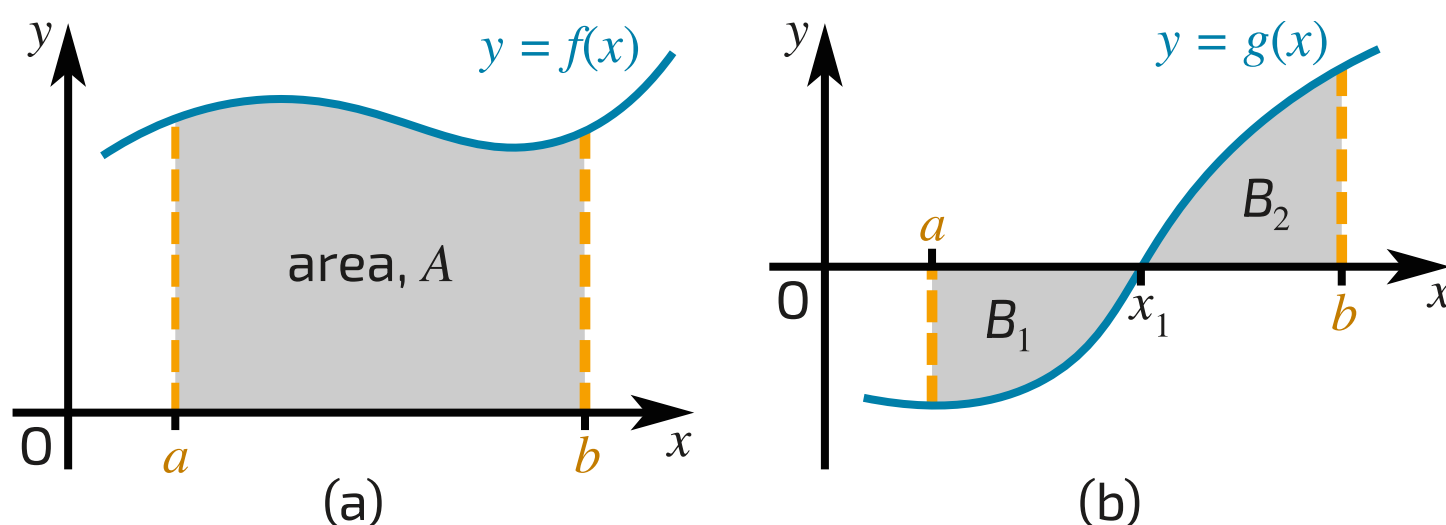


Figure 1: The area between a curve and the x -axis (a) for a function $y = f(x)$ that lies above the x -axis (b) for a function $g(x)$ that crosses the x -axis.

Integration can be used to find the under a curve.

In **Figure 1(a)**, the function $y = f(x)$ is the x -axis. The area of the region bounded by the curve $y = f(x)$, the x -axis, and the lines $x = a$ and $x = b$ is given by

$$\text{area, } A = \int_{\text{input}}^{\text{input}} f(x) \, dx$$

In **Figure 1(b)**, the function $y = g(x)$ crosses the x -axis. The total area enclosed between the curve and the x -axis between the lines $x = a$ and $x = b$ is $B_1 + B_2$. To calculate this area, we need to carry out separate integrals for the parts above and below the x -axis. The reason for this is that when a curve lies the x -axis, the indefinite integral gives the area multiplied by a minus sign.

$$\text{total area} = B_1 + B_2$$

$$\text{total area} = -\text{input} g(x) \, dx + \text{input} g(x) \, dx$$

Items:

a

b

$\int_a^{x_1}$

$\int_{x_1}^b$

above

area

below

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Functions from Differential Equations 2ii

Subject & topics: Maths Stage & difficulty: A Level P2

The gradient of a curve is given by $\frac{dy}{dx} = 6x - 4$. The curve passes through the distinct points $(2, 5)$ and $(p, 5)$.

Part A

Equation of curve

Find the equation of the curve.

The following symbols may be useful: x , y

Part B

Find p

Find the value of p .

The following symbols may be useful: p

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Functions from Differential Equations 1ii

Subject & topics: Maths **Stage & difficulty:** A Level P2

The gradient of a curve is given by $\frac{dy}{dx} = 12\sqrt{x}$. The curve passes through the point (4, 50). Find the equation of the curve.

The following symbols may be useful: c , x , y

Used with permission from UCLES, A Level, June 2012, Paper 4722, Question 2

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Equation of Curve

Pre-Uni Maths for Sciences K1.7

Subject & topics: Maths | Calculus | Integration

Stage & difficulty: A Level P1

Find the function $v(u)$ such that $\frac{dv}{du} = \frac{1}{3}u^{\frac{1}{3}} \left(1 - \frac{1}{u}\right)$ and $v(8) = -1$.

The following symbols may be useful: u , v

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Functions from Differential Equations 1i

Subject & topics: Maths **Stage & difficulty:** A Level P2

A curve has an equation which satisfies $\frac{dy}{dx} = kx(2x - 1)$ for all values of x . The point $P(2, 7)$ lies on the curve and the gradient of the curve at P is 9.

Part A

Find k

Find the value of the constant k . Give your answer as an improper fraction.

The following symbols may be useful: k



Part B

Equation of curve

Find the equation of the curve.

The following symbols may be useful: x, y

Used with permission from UCLES, A Level, January 2013, Paper 4722, Question 3

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Integrating to Find An Area 1

Subject & topics: Maths | Calculus | Integration **Stage & difficulty:** A Level P2

This question is about the use of integration to find the area of one or more regions between a curve and the x -axis.

Part A

Calculating the area under $y = x^2 + 1$

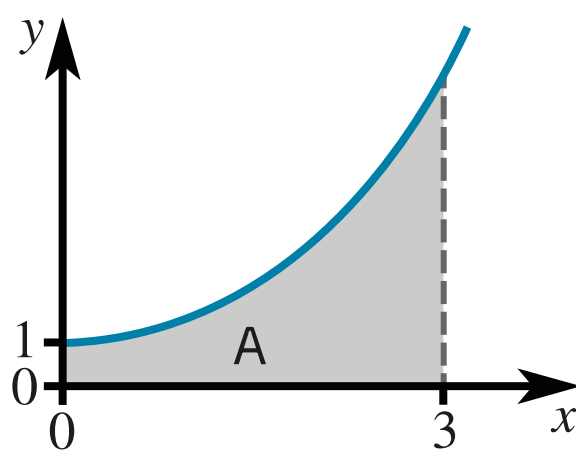


Figure 1: The graph of $y = x^2 + 1$

Figure 1 shows the curve $y = x^2 + 1$. The region between the curve and the x -axis, bounded by the lines $x = 0$ and $x = 3$, is labelled A. Calculate the area of A.

Part B

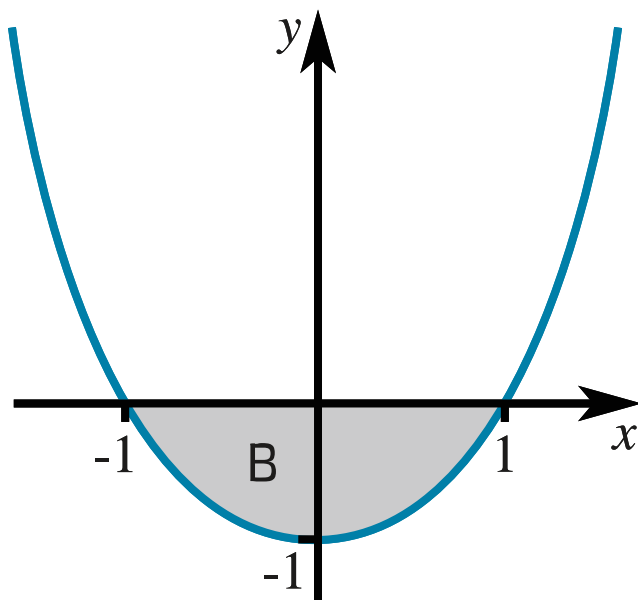
Calculating the area above $y = x^2 - 1$ 

Figure 2: The graph of $y = x^2 - 1$

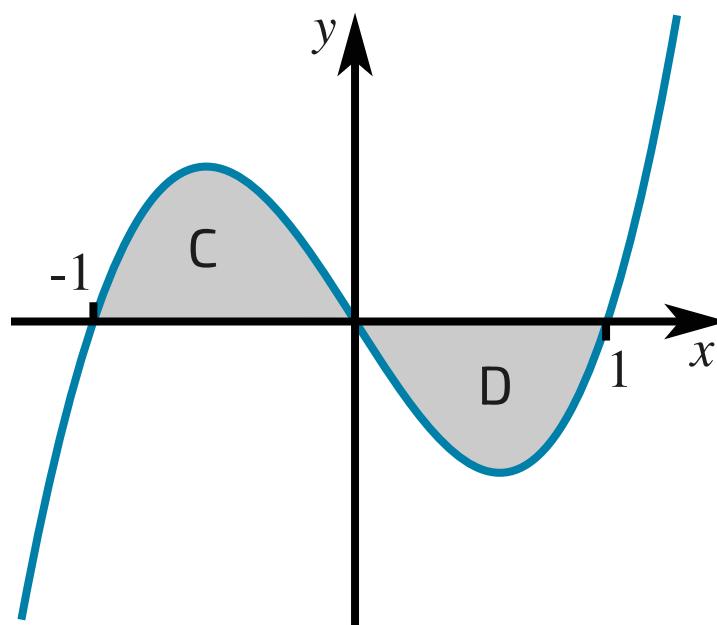
Figure 2 shows the curve $y = x^2 - 1$. The region between the curve and the x -axis, bounded by the lines $x = -1$ and $x = 1$, is labelled B. Calculate the area of B.

Part C

Integrating $5x(x^2 - 1)$

Calculate the value of the integral $\int_{-1}^1 5x(x^2 - 1) \, dx$.

Part D

The region enclosed between $y = 5x(x^2 - 1)$ and the x -axisFigure 3: The graph of $y = 5x(x^2 - 1)$

Use **Figure 3** to explain why $\int_{-1}^1 5x(x^2 - 1) dx$ does not give the total area of the regions enclosed by the curve $y = 5x(x^2 - 1)$ and the x -axis between $x = -1$ and $x = 1$. Drag and drop options into the spaces provided to complete your answer.

The curve $y = 5x(x^2 - 1)$ intercepts the x -axis at $x = -1$, $x = 0$ and $x = 1$. For $-1 < x < 0$ the curve is the x -axis. Hence, the value of $\int_{-1}^0 5x(x^2 - 1) dx$ is and equal to the area of region . However, for $0 < x < 1$ the curve is the x -axis. Hence, the value of the integral $\int_0^1 5x(x^2 - 1) dx$ is . The area of the region labelled is given by $-\int_0^1 5x(x^2 - 1) dx$.

The total area of the shaded regions is $\int_{-1}^0 5x(x^2 - 1) dx - \int_0^1 5x(x^2 - 1) dx$. This is not the same as $\int_{-1}^1 5x(x^2 - 1) dx$, which has a value of 0 as the curve has symmetry and the contributions from the parts above and below the x -axis cancel out exactly.

Items:

D C positive below above negative reflective rotational

Part E

Calculating an area for $y = 5x(x^2 - 1)$

Calculate the area of the region enclosed between the curve $y = 5x(x^2 - 1)$ and the x -axis for x between -1 and 1 .

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Integration and Area 1ii

Subject & topics: Maths **Stage & difficulty:** A Level P2

Figure 1 shows part of the curve $y = x^2 - 3x$ and the line $x = 5$.

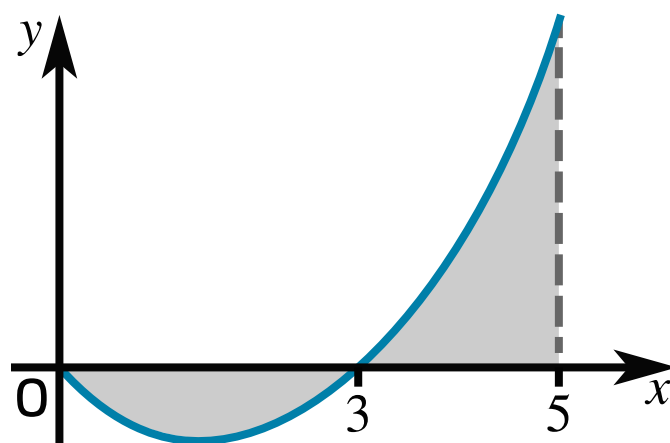


Figure 1: A graph of $y = x^2 - 3x$ and the line $x = 5$

Part A

Area of shaded regions

Which of the following expressions gives the total area of the regions shaded in **Figure 1**? Explain your choice.

- ☐ $\int_3^5 (x^2 - 3x) \, dx - \int_0^3 (x^2 - 3x) \, dx$
- ☐ $\int_0^3 (x^2 - 3x) \, dx + \int_3^5 (x^2 - 3x) \, dx$
- ☐ $\int_0^5 (x^2 - 3x) \, dx$

Part B
Find area

Use integration to find the total area of the shaded regions. Give your answer to 3 significant figures.

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Area Under a Curve 1

Pre-Uni Maths for Sciences K2.7

Subject & topics: Maths | Calculus | Integration Stage & difficulty: A Level P2

A graph of the functions $y = (x - 2)(x + 1)$ and $y = x + 1$ is shown in **Figure 1**. Find the areas of the shaded regions labelled A and B. A is the region between P and Q enclosed by the curve $y = (x - 2)(x + 1)$ and the x -axis; B is the region between Q and R below the curve $y = (x - 2)(x + 1)$ and above the x -axis.

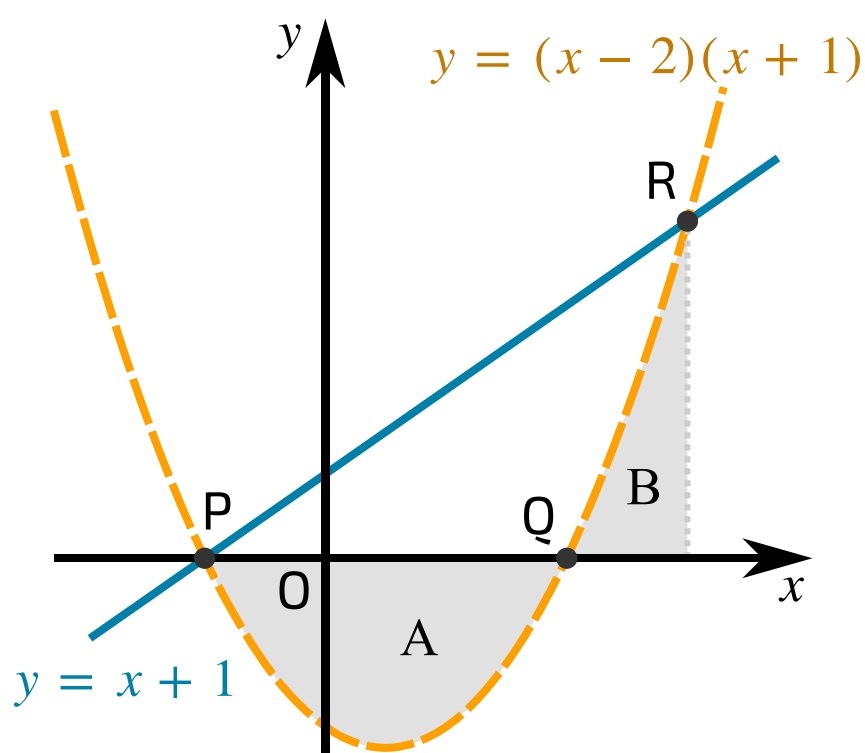


Figure 1: A graph of the functions $y = (x - 2)(x + 1)$ and $y = x + 1$. A is the region between P and Q enclosed by the curve $y = (x - 2)(x + 1)$ and the x -axis; B is the region between Q and R below the curve $y = (x - 2)(x + 1)$ and above the x -axis.

Part A

Region A

Find the area of the region A. Give your answer in the form of an improper fraction.

Part B
Region B

Find the area of the region B. Give your answer in the form of an improper fraction.

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Integration and Area 3i

Subject & topics: Maths **Stage & difficulty:** A Level P2

Figure 1 shows the graph of $y = 1 - 3x^{-\frac{1}{2}}$.

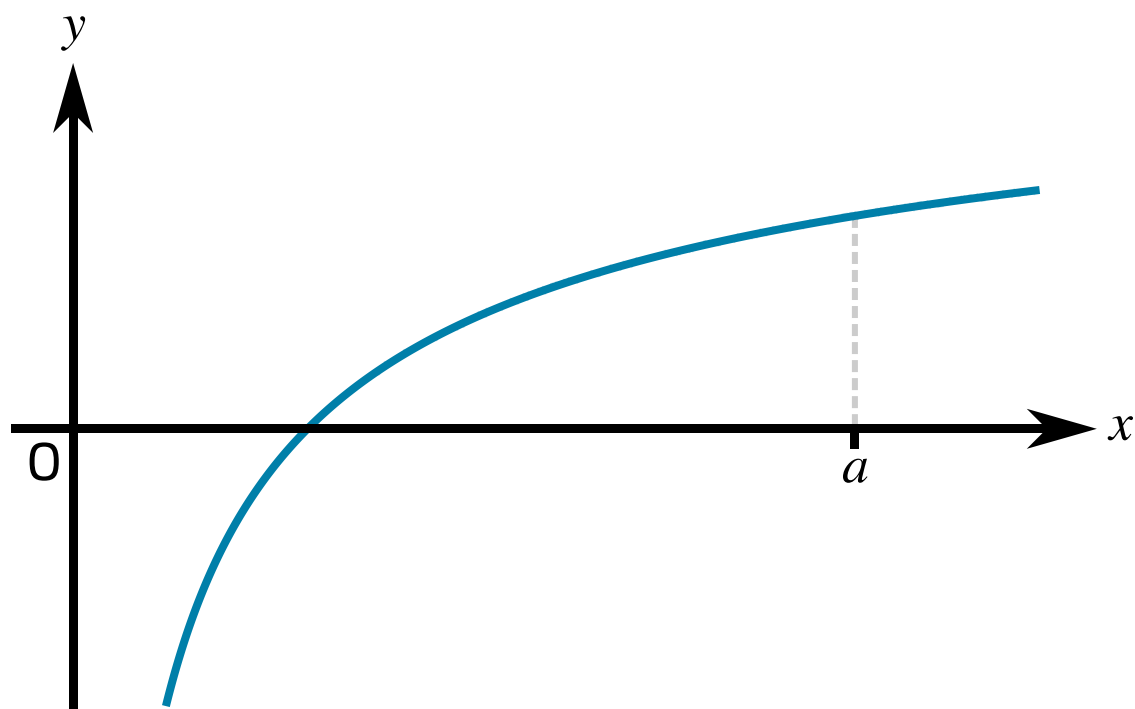


Figure 1: Graph of $y = 1 - 3x^{-\frac{1}{2}}$.

Part A

Find intersection with x -axis

Find the x -coordinate of the intersection of that curve with the x -axis.

The following symbols may be useful: x

Part B**Find a**

The region enclosed by the curve, the x -axis and the line $x = a$ (where $a > 9$) has an area equal to 4 square units, find the value of a .

The following symbols may be useful: a

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Area Under a Curve 3

Pre-Uni Maths for Sciences K2.9

Subject & topics: Maths | Calculus | Integration Stage & difficulty: A Level C2

A graph of the functions $y = \frac{1}{2\sqrt{x}}$ and $y = 2x\sqrt{x}$ for $x \geq 0$ is shown in **Figure 1**. Find the area of the shaded region OPQR.

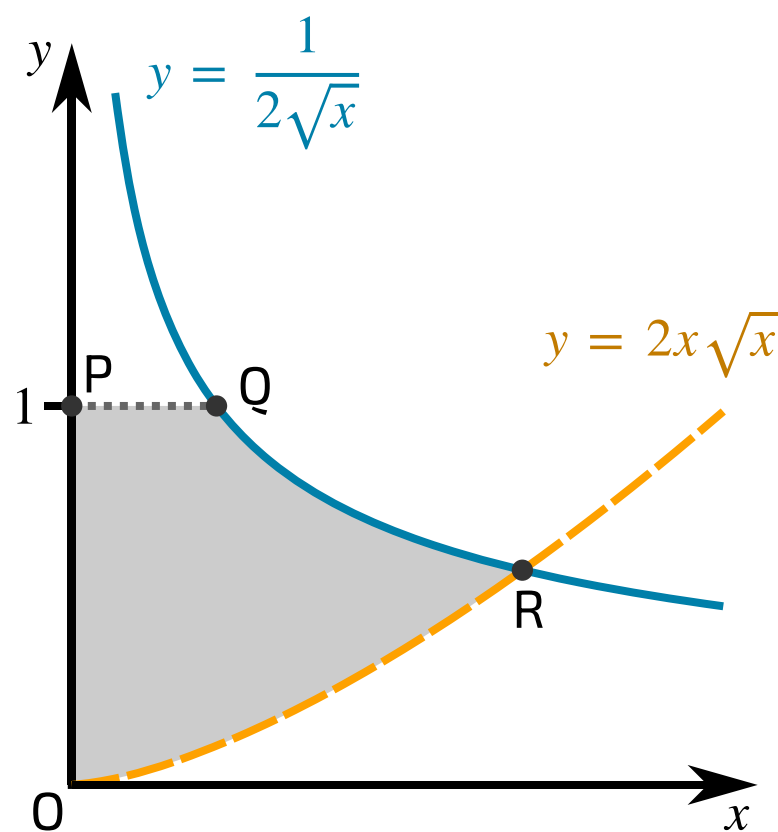


Figure 1: A graph of the functions $y = \frac{1}{2\sqrt{x}}$ and $y = 2x\sqrt{x}$ for $x \geq 0$. The shaded region OPQR is bounded by the line $x = 0$, the line $y = 1$, the curve $y = \frac{1}{2\sqrt{x}}$ and the curve $y = 2x\sqrt{x}$.

Part A

The x -coordinate of QDeduce the x -coordinate of the point Q.

Part B

The x -coordinate of R

Find the x -coordinate of the point R.

Part C

The area of OPQR

Find the area of the shaded region OPQR, giving your answer in an exact form.

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